HRF (HTTP Request Filtering): A New Detection Mechanism of EDoS Attack on Cloud

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***Abstract*: This paper studies the impact of EDoS attacks with proposed detection mechanism of Web Application Firewall (WAF). The HTTP Request Filtering (HRF) technique is proposed as a detection mechanism and implemented using predefined rules and conditions visa-vis using threshold , throughput and inter arrival time of the requests on WAF . Evaluated analytical queuing model for the proposed defense mechanism and also explains the work flow. This analytical queuing model verified through experimental test-bed to study the implemented WAF performance on the EDoS attacks in Cloud Computing (CC) environment. Evaluated performance metrics are response time, utilization, and throughput and these metrics have used for cost analysis. These results are presented in tables as well as graphs and conclusions are drawn from the study.**

Keywords: Web application firewall; HTTP request filtering, queuing model; EDoS attack; conditions.

I. INTRODUCTION

CC Security (CCS) is a combination of infrastructural, logical and informational security issues in cloud service deployment models in the CC Environment (CCE) [1]. It also explains about how these services are processed to provide the information and resource security for customers. As per the [2] CC is a technology for sharing pool of on-demand computing, network, physical and storage resources on pay-as-you-go-basis. The public cloud CCS guidelines and issues are explained in [1]. Security issues of cloud provide different levels of mechanism within the network resource to provide protection for cloud oriented denial of services like websites and web applications. The CCS mainly concentrates on protection from DDoS, resource availability and security for stored data. A DDoS attack goal is overwhelm web application servers so it cannot respond for the legitimate clients. If a DDoS attack activity is successful, then it leads to web server unavailability for certain time. This can leads to loss of revenue and customer.

The Software-as-a-Service (SaaS) oriented applications uses infrastructure or hardware as main resources to deploy web applications on the cloud. Here resource and web service availability are the main challenging factors. The sophisticated attackers generate http load spike on the hosted web application to create unavailability of web services and resources for the legitimate users. This is called as Distributed-Denial-of-Service (DDoS) attack [3]. Rapid elasticity is one of the essential characteristics of CC and it shows solution to the DDoS at the same time it introduced another type of assults, that is Economical-Denial-of-Sustainability (EDoS) attack. The EDoS attackers main goal is economic loss of the cloud vendor, thus they increase the load on web application to utilize the resources and network [4] [5] [6].

The EDoS attack detection is the challenging task of CCS environment. Most of the existing detection systems are proposed in literature survey with shields, fuzzy entropy, Ads and frameworks in the same way [7-13]. But they only considered firewalls of resources and pre-active approaches. Although the sophisticated attackers concentrate on the web application and related integrated services, for example many number of US based banking and e-comers web sites are under the victim activity and customers are suffered for hours [14]. Similarly UK based HDFC Bank [15] and Bitcoin websites have targeted by the attackers like banking websites [16] [17]. Kubra Kalkan [56] proposed a Statistical based filtering mechanism for DDoS attacks but it is a proactive approach and independent model. There are many DDoS mitigation mechanisms but most of them are not suitable for EDoS mitigation [18]. To fulfill the gap there is a necessity to research on HTTP load spike filtering and WAF filtering mechanism to mitigate EDoS attacks.

This paper proposes HRF mechanism in WAF. In this framework customers are categorized into three types: white list, block list and unknown users. Implementation of the mechanism following three fold HTTP load filtering conditions, one is threshold size of the HTTP request and second is throughput of the source IP address and finally inter arrival time of the source IP address. Considered WAF based web application hosting architecture and evaluated analytical model for proposed filtering mechanism. Implemented analytical illustration for performance of the architecture using two scenarios: with filtering mechanism and without filtering mechanism. The analytical results are compared with experimental results and finally evaluated cost function of the cloud.

The remaining part of this paper is structured as follows. Section 2 explains the survey behind the filtering mechanism with the gained knowledge in the literature study. Section 3 illustrates about web application firewall based architecture and also explained the work flow of FM using analytical model. Performance evaluation of architecture explains in section 4. The experiential evaluation has been discussed in Section 6. Obtained analytical and experimental results are compared and discussed in section 5 and finally conclusions are presented.

II. LITERATURE SURVEY

This survey section briefly discussed about that are in the previous systems to abate the effects of EDoS attacks, moreover which become the motivation for the proposed filtering mechanism and considers both EDoS and application layer DDOS detection techniques.

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Before going to EDoS attack mitigation a small review on DDoS attack mitigation because EDoS is the transformation of the DDoS. Few authors are founded Inter Arrival Time (IAT) is playing the important role to mitigate the DDoS including EDoS [19], thus our proposed filtering mechanism also considered this as a one of the condition to detect the EDoS attack.

Amit Praseed et al. [3], these authors explained about the DDoS attacks execution and reviewed different detection and mitigation mechanisms of network application layer in the CC. The authors are proposed detection of DDoS attack in the application layer using single class web application in the cloud. [20] In this paper authors are discussed different attacks on web services and also provided SOAP protocol execution in the DDoS attacks. At present researchers are focused on the HTTP oriented DDoS attacks. Zargar et al. [21] and Aamir & Zaidi [22] both explained network and application layer attacks.

Compare to the previous DDoS, there is a need to detect and mitigate the dedicated EDoS attacks. Thus, this paper discuss next about the existing mitigation and detection techniques of EDoS. In [48] [23] the authors are surveyed in the EDoS attack mitigation. They are concludes signature-based detection techniques are faced problem by the unknown DDoS and existing techniques have drawbacks. Two surveys highlighting EDoS attacks mitigation adequately require many robust techniques.

The EDoS attack mainly focus on the utilization of the cloud by using elasticity property to get excess cost for the provider's. [8] , in this paper the authors are proposed a novel reactive approach to reject EDoS attacks while considering into the existing drawbacks. The EDoS-ADS is used to recognize genuine users from the pool of request through NAT based network.

Amazon introduced CloudWatch [24] to minimize the EDoS attack risk by using auto scaling property and along monitoring. CloudWatch was configured with min and max elasticity limits and thus reduce the EDoS impact. It will generate the trigger when elasticity exceeds max elasticity and holds the service. Thus, the genuine customers also unable to get the cloud services until cloud provider decision.

Self-verifying Proof of Work (sPoW) is a technique and it shows the solution through crypto-puzzles [25]. Asymmetric power consumption problem is one of the faults in this paper. The customers are must solve the client puzzles thus it leads to false positives. The Gligor [26] presents that the client puzzles are behaving like work proof to provide assurance service to the customers. Similarly many authors are presented sPoW techniques in their work [27], [28], [29].

Al-Haidari et al. [13] proposed EDoS-Shield as a mitigation technique. This architecture contains 2 states. The first state is a virtual-firewall along with white liast and black lists and the second state is a V-Node. V-Node was implemented using GTT [30]. By using GTT the cloud provider send the visual tests to customers. The GTT takes decision based on the rest results they are legitimate or not. Subsequently updates white and block list. Finally, the EDoS-Shield uses the firewall to check the load spike. This leads some significant difference on the response time of the legitimate users to take the service on the cloud. The authors [31] are enhanced EDoS Shield to rectify the spoofed IP addresses problem in [13]. They are included Time To Live (TTL) value to source IP and include unmatched TTL values to the white and black lists to take decision. This also has limitations like EDoS Shield [13]. The attackers are have a possibilities to change the existed TTL, so it is not always right [47].

EDoS Armor [32] mitigation technique is proposed by Masood et al. It has dual detection system; admission and congestion control. It has more short comes like elasticity property as the admission control limits and the parallel customer requests for web services. Baig et al. [33] introduced another EDoS mitigation technique. It contains three stages such as virtual firewall (VF), an ELB with elastic cloud and a database. This technique is used CPU utilization as a main resource to mitigate the attack. The proposed method has some limitations such as turning test and it blocks total NAT based network if it not genuine.

III. WAF ARCHITECTURE FOR WEB APPLICATION HOSTIING

A. AMAZON WAF ARCHITECTURE

In our study this paper focused on the usage of WAF and implemented HRF as a workflow of WAF and it is used to detect EDoS attacks on web services of the cloud. This cloud adopted a single class SaaS type application on the centralized datacenter. The malicious users are utilizing this opportunity to consume the cloud resources as well as to create unavailability environment for legitimate users while they offer the web service. This process increases the vendors cloud maintenance cost.

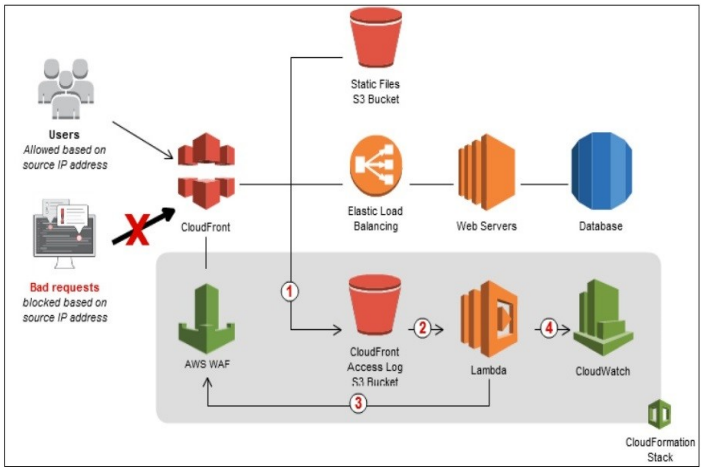


Fig 1: WAF deployment for web application services in AWS

Amazon WAF based architecture to be identified for evaluate the proposed model and conduct experiment for this study and it depicts in Figure 1 [34]. Main components of this architecture are Web Application Firewall (WAF), CloudFront (CF), S3, Lamda, CloudWatch, Elastic Load Balancer (ELB), Elastic Cloud and Database Server.

The WAF main goal is to detect the risk traffic and protect the web service against the attackers like DDoS attacks, HTTP flooding attacks etc. the firewall is positioned before the web application server and it configures with certain security rules, conditions, web access control list and actions to detect and mitigate attackers targeted at web applications. Rules are nothing but conditions, when all rules are satisfied then only the rule active. Conditions are used to filter the suspected traffic. The WAF first of all focuses on the URL IP, any query, HTTP header, methods and request source IP address. Web ALC

consist many rules along with an action. These actions are like the rules and are the part of the conditions. An action is similar to the counter and gateway.

The elastic cloud contains web application servers and they run simultaneously in the centralized datacenters, each server has the capacity to process client requests for servicing [35]. The elasticity property can be managed by auto scaling group and is depend on the parameter such as average CPU utilization of the configured cloud [36].

B. PROPOSED HRF MECHNISM WORKFLOW IN WAF

A HRF is a mechanism to detect the EDoS attacks. Figure 2 shows the queuing model based filtering mechanism. This mechanism consists of three phases the Request Filter (RF), Network Analyzer (NA). The Request Analyzer (RA) receives three types of arrival rates legitimate user (λ_l), malicious user (λ_m) and unknown users (λ_{un}). The RF was predefined with rules and storage register. The storage register contain two types of registers one is white list for legitimate source IP addresses and other one is for malicious source IP addresses. The RF checks the source address and compare with storage register data so as to identify the request source is white list or block list and it comes to conclusion about its processing, if the source address is in block list its process to RA else its process to NA. The NA handles legitimate users and unknown users and it checks the size of the request compared with predefined web application threshold value. The legitimate requests exceeds the threshold value then the request sent to RA else it will be serviced by the internal load balancer. The unknown users request was processed by the RA.

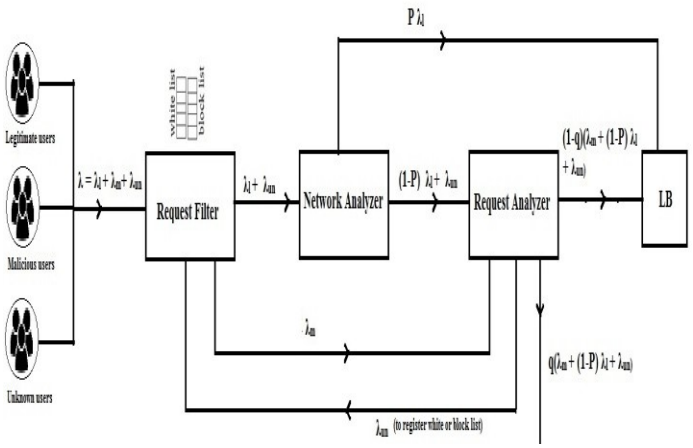


Figure 2: HRF Mechanism workflow

The RA will analyze the throughput of the arrivals source IP address. If it exceeds the predefined request rate in terms of seconds then it detects as DoS attack and takes the responsibility to block the request and generate trigger to register particular IP address in the storage register block list else it will be serviced by the internal load balancer. The unknown user requests are travelled by the two phases and the RA will take decision to register in the white list or block list.

The RFM modeled as M/M/1 queuing model with Poisson arrivals and exponential service. Total arrival rate of the system is $\lambda = \lambda_l + \lambda_m + \lambda_{un}$. The request filter distributes requests λ_l and λ_{un} to NA and λ_m to RA. The NA will be sent to load balancer with P probability and RA with (1-P) probability. The request analyzer arrival rate is $\lambda_m + (1\text{-}P)\ \lambda_l + \lambda_{un}$. Arrival rate of the requests serviced by the internal load balancer is $P\lambda_l + (1\text{-}q)(\ \lambda_m + (1\text{-}P)\ \lambda_l)$ and it distribute to servers on the elastic cloud.

C. HRF MECHANISM FLOWCHART

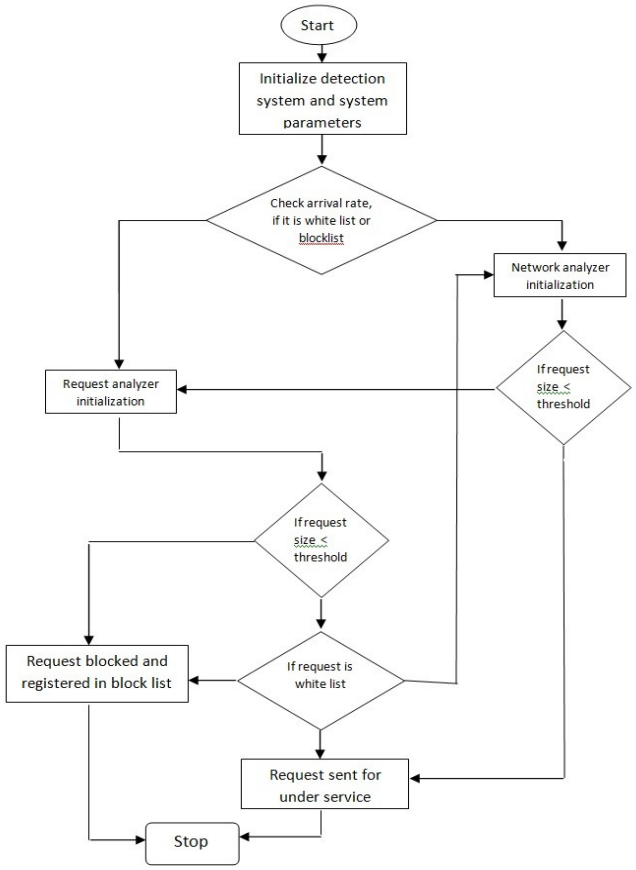


Figure 3: Flowchart of Filtering System

IV. PERFORMANCE EVALUATION OF THE WAF ARCHITECTURE

An analytical model is evaluated to show the performance of the filtering mechanism in the WAF that is applicable in mitigating the EDoS attacks. Figure 4 represents successive phases of queuing model for WAF architecture presented in figure 1. The queuing model contains 3 phases, filtering mechanism, computing and storage. The FM and computing phases are represented by the group of parallel VM instances. The arrival rate follows Poisson distribution, as reported in [37] [38][39][40]. The analytical model analyze the dynamic behavior of the system to study the performance metrics include latency of the http request, elastic cloud utilization , total number of servers running on the cloud, throughput and setup and maintenance cost.

In the queuing model architecture FM and computing services contain load balancer. These load balancers are modeled as M/M/1 queuing model and distribute the load spike in the midst of available instances in FM and cloud service phases, thus each VM has receive same probability load. In FM and computing tire are modeled as parallel M/M/1 queuing models [35] [49] [50]. As per the Amazon the storage supports vertical scaling, thus this service modeled as single M/M/1. The queuing model shown in figure 4 is basically M/G/1 with poisson arrivals

and distributed service times. The request execution met by WAF, web application server and storage.

The http proxy distributed evenly by the LB in the midst of available instances in FM and Computing, then the transition probability will be equal to 1/M and 1/N for all the FM and computing instances, where M and N are the total number of computing instances. In the elastic cloud concept all the bounded VM's have the same utilization power capacity shown in the figure 4, thus $\alpha_i = \alpha$ and $\mu_i = \mu$. The FM and computing tire arrival rate of each VM follows $\lambda i = \lambda/M$ and computing each instance is $\lambda_i = \lambda_l/N$. Then the average resource utilization in both tires can be calculated as:

$$U_{FM} = \frac{\lambda_l+\lambda_m+\lambda_{un}}{M\alpha} \qquad U_{EC} = \frac{\lambda_l}{N\mu} \qquad (1)$$

The FM and Computing phases have open queue network model of M and N parallel queues. As per [30] the latency or response time of the request can be calculated as below. The LB distribute the load in the midst of M and N running instances then the probability routing matrix has P_i and P_j , that are equal to 1/M and 1/N, where P_i and P_j are the routing probability to the ith instance in FM and j^{th} instance in the EC. The total received load in the FM and EC are: $\Lambda_i = \lambda/M$ and $\Lambda_j = \lambda/N$.

Based on M/M/1 queuing theory the total mean delay of a load in VM i and j in the FM and EC can be described to be: $\bar{T} = \frac{1}{\alpha-\Lambda i}$ and $\bar{T} = \frac{1}{\alpha-\Lambda j}$

As per the Little's formula the network mean delay is:

$$\bar{T} = \frac{1}{\lambda}\sum_{i=1}^{M}\frac{\Lambda i}{\alpha-\Lambda i} \text{ and } \bar{T} = \frac{1}{\lambda l}\sum_{i=1}^{M}\frac{\Lambda j}{\mu-\Lambda j}$$

Assuming that based on rapid elasticity nature there is no delay to enable and bind new VM in the EC group, the average latency of the load request of queuing model of FM and EC can calculate as:

$$R_{FM} = \frac{M}{M\alpha-\lambda} \text{ and } R_{EC} = \frac{N}{N\mu-\lambda_l} \qquad (2)$$

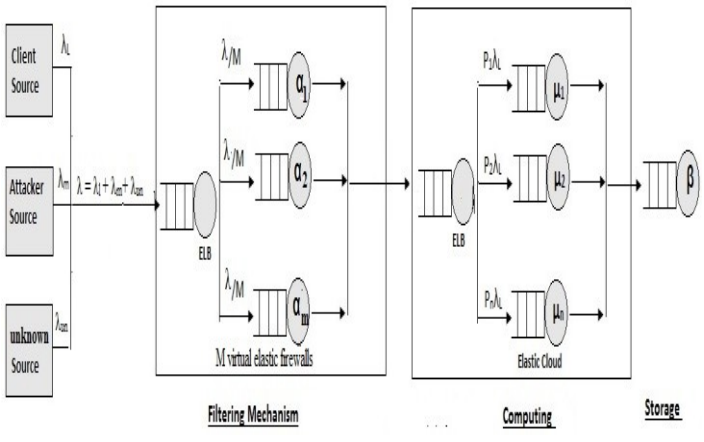


Fig 4: Queuing model for web application firewall based web application

The Amazon AWS used RDS servers as storage tier to manage or store relational database. It has several features live vertical scaling, monitoring, replication, point-in-time backup or recovery of the database server. RDS server not like elastic server and it supports vertically scalable on the database tier, and it unable to add database. Thus, storage tire in the architecture could be modeled as a single M/M/1 queue [54, 55]. The utilization and response time of the storage can be calculated using following formula:

$$U_{Sto} = \frac{\lambda_l}{\beta} \text{ and } R_{Sto} = \frac{1}{\beta-\lambda_l} \qquad (3)$$

This paper ignores the delay because of focus on the performance of filtering mechanism. From equations (1), (2) and (3) The total average response time is calculated using following formula $R = R_{FM} + R_{EC} + R_{Sto}$.

$$R = \frac{M}{M\alpha-\lambda} + \frac{N}{N\mu-\lambda_l} + \frac{1}{\beta-\lambda_l} \qquad (4)$$

The utilization U of computing resources of filtering mechanism, computing tire and storage tire is calculated using following formula $U = U_{FM} + U_{EC} + U_{Sto}$.

$$U = \frac{\lambda_l+\lambda_m+\lambda_{un}}{M\alpha} + \frac{\lambda_l}{N\mu} + \frac{\lambda_l}{\beta} \qquad (5)$$

The legitimate throughput of the elastic cloud can be calculated using Little's Law. The legitimate arrival rate λ_l is evenly split into N number of running instances, thus each VM arrival rate is λ_l/N and the service rate is μ. Then the throughput of the S running instances is $N \times (\mu \times \rho) = N (\mu \times (\lambda_l/ N\mu)) = \lambda_l$.

The cost of the cloud can be impacted due to EDoS attack [4][5]. The cloud adopter bill effected based on the usage of network, computing and storage resources utilized. The cost equation can be calculated as described below:

$$Cost=(P_{nw} \times (\lambda_l + \lambda_m + \lambda_{un}) + (P_{fm} \times M + P_{ecomp} \times N) \times U + P_{sto} \times \lambda_l) \times T \qquad (6)$$

Where P_{nw} is price of the bandwidth, P_{fm} is price of computing instances in the filtering mechanism, P_{ecomp} is price of computing instances in the elastic cloud and P_{sto} is database instance cost.

This paper for the numerical illustration assumed based on the size of the web application capacity of the micro virtual machine is 100 requests per second [43]. This elastic cloud initially started with 5 servers. In the auto scaling provisioning overhead considered as 55.4 s, and it was taken from Islam et al. [44]. In our assumptions $1/\beta = 20$ ms, legitimate users traffic 400 requests per second, attacker traffic varying from 200 to 2000 requests per second . Total cost of the elastic cloud datacenter can be calculated using equation (6). As per the amazon aws price of the micro instances is $0.115, RDS server price is $0.115, base cost of $0.01 per GB in/out information exchanged dependent on the revealed costs of Internet information exchange "in" and "out" of Amazon EC2 [45].

V. EXPERIMENTAL ANALYSIS

This paper conducted an experimental model to enhance the analytical model of filtering mechanism. This simulation model contains two scenarios. In the first scenario, the web application

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protected with proposed detection and mitigation mechanism to reduce the EDoS attack risk. In second scenario there is no firewall and it considered as optimal case. The entire experiment has been conducted on the Amazon AWS.

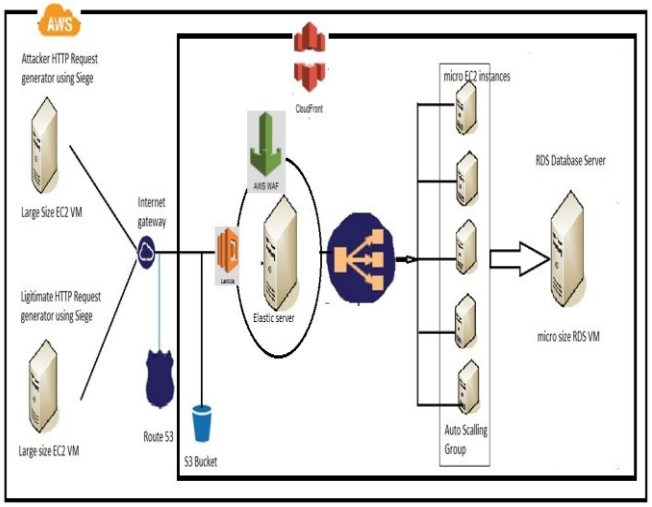


Fig 5: Experimental model for web application firewall based web application

Figure 5 shows the experimental setup design to host the firewall based web application on AWS. Single class software as a service based web application hosted in the single server. This server was configured with auto scaling, elastic load balance and RDS database server. The web application firewall was configured with IP match condition, geo match condition, string match condition, regex condition and sql injection match condition. In addition it was configured with our proposed filtering mechanism conditions in the rules list. According to wool [46] maximum size of the rule set of the virtual firewall is 2671. The AWS CloudFormation is act as a template to create or configure the conditions, rules, and a web ACL, this CloudFormation associate with CloudFront distribution to activate the AWS WAF. The S3 bucket is added to this server to store the access logs. This bucket acts as a white list and block list register. Finally registered domain name using Route 53 and associated this CloudFront. Create a CloudWatch dash board to monitor the web application.

Three large EC2 instances created to generate the HTTP proxy as legitimate, malicious and unknown users. This traffic received by the CloudFront behalf of web application through Route 53. The CloudFront sends access logs to S3 bucket. Every new request is stored in S3 and this request was checked by the Lamda. This Lamda checks threashold and period time of request and compare with WAF rules, if it is malicious then it generates trigger for this request to store in the block list. If it is legitimate then the request will store in the white list. The Lamda sends the request results to CloudWatch to monitor the attacker activity. White list logs are sends to receive web service.

The entire experiment was conducted in the region of N.Virginia. The hosted web page size is 580 bytes; thus the micro instance achieves 100 % utilization when the VM receives 100 req/sec in the micro instance. In this experimental model the legitimate user's arrival rate fixed with 400 Req/s. The malicious and unknown users' traffic was varied from 200 to 2000 req/s. Finally monitored various performance metrics average cloud resource utilization, End to end latency, legitimate throughput, maximum number of instances and Cost.

VI. RESULTS AND DISCUSSIONS

This paper has considered four scenarios for obtained results. Scenario one is performance of the analytical model without HRF mechanism, scenario two is experimental model without HRF mechanism, scenario three is analytical model with HRF mechanism and finally forth scenario is experimental model with HRF mechanism. The performance metrics are latency or response time, cloud resource utilization, throughput of the legitimate users and setup and maintenance cost of the setup are evaluated in the four scenarios. This experiment conducted by using different attack rates to identify EDoS attacks effect on the targeted cloud and also proposed FM to mitigating the attacks impact.

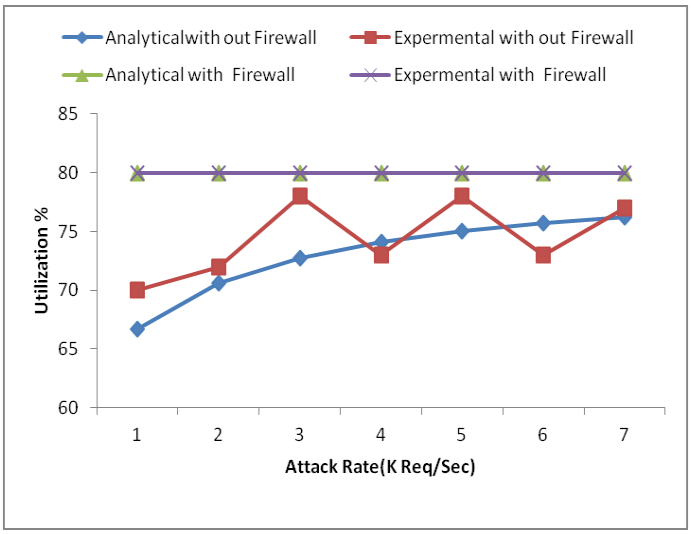


Fig 7: Impact of the cloud utilization when attack rate increases

From the Figure 7 the analytical values of without HRF scenario are increases when the attack rate also increases and number of servers also increases. The experimental values are oscillated within the range of 70 to 75. In HRF scenario the analytical and experimental results are follows the same trend with almost all close to each other and number of servers are fixed with initial setup servers.

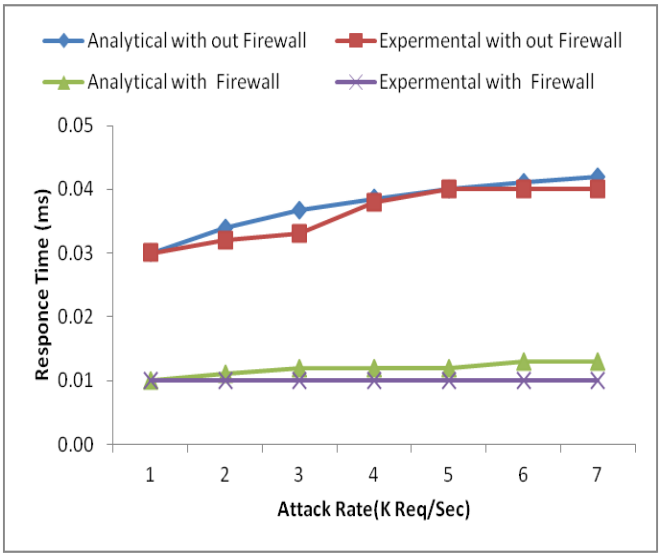


Fig 8: Impact of the response time when attack rate increases

Figure 8 shows the obtained results of response time from both analytical and experimental. In the both firewall and without firewall cases there is no significant difference between

the analytical and experimental. In addition, the results show considerable degradation effects on the cloud service performance in terms of the response time when the attack rate increases and without using mitigation technique.

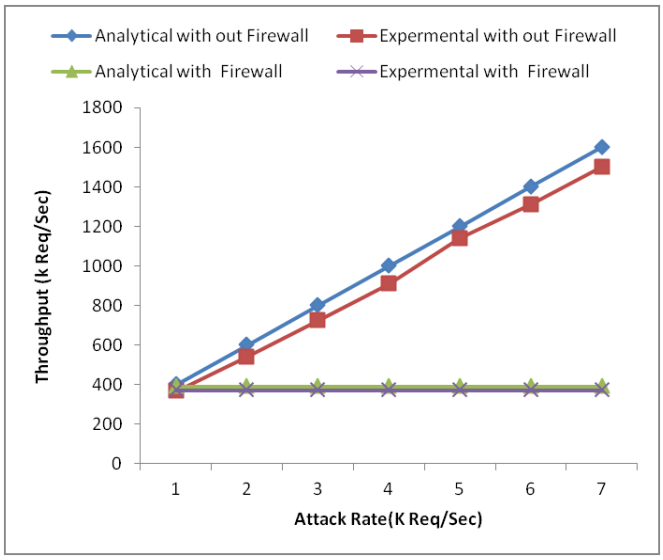


Fig 9: Impact of the throughput when the attack rate increases

The Figure 9 depicts the EDoS impact on the throughput parameter. In the without firewall case the throughput of the analytical and experimental model goes high when the attack rate increases and more significant for the large values of attack arrival rate. In the mitigation scenario the analytical and experimental values goes constant and there is no significance difference between them. In the mitigation scenario the legitimate throughput are under service and the attacker throughput was blocked by the firewall.

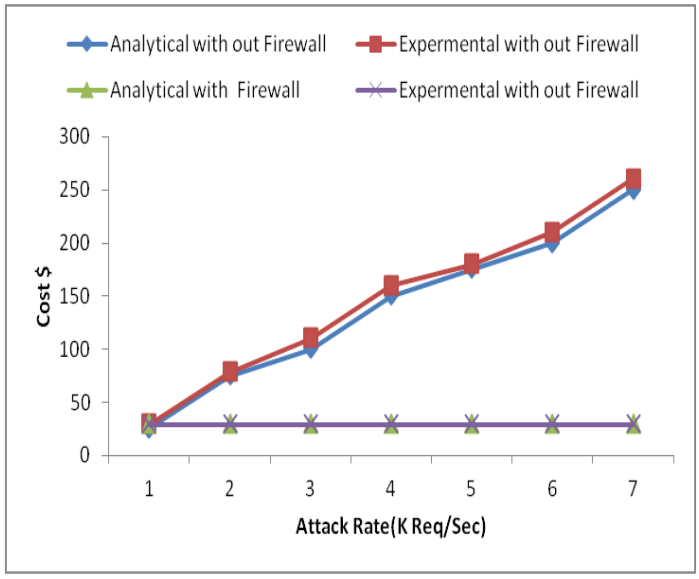


Fig 10: Impact of the cost when the attack rate increases

Figure 10 shows in the without firewall case when the attack rate increase the corresponding system cost also increases. In the mitigation scenario the analytical and experimental values goes constant and there is no significance difference between them. The firewall drops the attackers and the system cost is constant for various arrack rates.

VII. CONCLUSIONS

This paper investigates the proposed HRF mechanism performance for EDoS attack mitigation. The evaluated HRF analytical model verified with the experimental study. These analytical and experimental results conclude that the HRF mechanism is better than without HRF mechanism when calculate the performance metrics like utilization, throughput of the legitimate users, response time and cost metric. In without HRF scenario when the attacker throughput increases the corresponding total number of running servers are increased, thus the utilization of the cloud goes high. Due to the elasticity property there is no significant effect on the latency. In addition cost of the cloud increases and it leads to economical loss of the business or cloud vendors. In second scenario the HRF mechanism detects the attacker's activity and drops the malicious requests. So performance metrics goes to constant. There is no significant difference between analytical and experimental models. To implement the HRF mechanism for different types of firewalls is a future scope of the work.

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